Tracing Sparse Causal Features in an Open Language Model

A Small-Scale Reproducibility Study of Attribution Graphs

MPhil in Data Intensive Science

Project 28: Mechanistic Interpretability of Open Source LLMs

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Abstract

Anthropic's *On the Biology of a Large Language Model* uses sparse features, attribution graphs and causal interventions to test hypotheses about language-model mechanisms. Its Claude 3.5 Haiku analysis employs a cross-layer transcoder (CLT) as a local replacement for multilayer-perceptron (MLP) outputs. This study investigates whether that workflow transfers to the open-weight, four-billion-parameter `Qwen3-4B-Instruct-2507` under an independent small-scale design. Behaviour-specific sparse autoencoders (SAEs) are trained on captured MLP outputs from synthetic prompt sets and evaluated through attribution graphs, inhibition and swap-in interventions. The behaviours are two-digit addition with emphasis on carry propagation, capital-city reasoning, and physics-related factual reasoning with SI-unit prompts.

The analysis separated reconstruction, decodability and causal control through successive falsification tests. Initial ReLU SAEs reconstructed held-out activations well, with mean validation fraction of variance explained (FVE) between 0.907 and 0.959, but activated 2,167–3,545 of 8,192 latents on average. These diagnostics established that high reconstruction fidelity coexisted with insufficiently sparse codes. Broad patches confirmed that relevant causal information was present.

Initial graph-derived and TopK arithmetic panels exhibited no carry specificity under matched no-carry controls. A subsequent output-digit-balanced analysis localised carry information to late MLP outputs, and inhibiting a frozen 20-feature panel produced a small carry-selective effect that replicated on 32 intervention-untouched cases (paired effect -0.086 logits; bootstrap 95% interval $[-0.137, -0.039]$). No answer token flipped. Capital-city swaps remained weak for both low- and high-confidence prompts.

The strongest effect occurred in the SI-unit TopK follow-up. The selected $k = 128$ SAE retained mean validation FVE 0.936 while reducing mean activity to 99.8 latents. Features were ranked on eight discovery systems and a frozen ten-feature panel was tested on 16 disjoint confirmation systems. Swapping force-source values into energy targets increased the newtons-minus-joules logit gap by 1.234 logits (bootstrap 95% interval $[1.148, 1.332]$), whereas a matched mass-source control changed it by -0.090 ($[-0.129, -0.051]$). The paired specificity effect was 1.324 logits ($[1.223, 1.430]$) and had the predicted sign in all 16 cases. Joules nevertheless remained the top prediction. The evidence supports compact, distributed carry- and force-associated causal panels, not complete circuits, monosemantic variables or answer flips.

Contents

1 Introduction

Large language models can produce correct outputs without revealing the internal computations that generated them. Mechanistic interpretability seeks causal accounts of those computations in terms of internal representations and their interactions. A central obstacle is superposition as described by Elhage et al.: a model may encode more features than it has individual neurons, so a neuron need not correspond to one human-interpretable concept [5]. Sparse autoencoders (SAEs) attempt to recover a more useful basis by reconstructing model activations from a small number of active learned directions [3, 4].

Lindsey et al. applied this programme at scale in *On the Biology of a Large Language Model* [1]. Their cross-layer transcoder (CLT) approximates Claude 3.5 Haiku's MLP outputs using sparse features that may align with interpretable concepts. Attribution graphs trace locally important paths from prompt tokens through those features to output logits, yielding hypotheses about mechanisms such as addition, multi-step thinking, and physics-based reasoning. The companion methods study defines an attribution graph as a local replacement model and hypothesis generator [2]. Causal intervention in the underlying model therefore provides the criterion for evaluating graph-derived hypotheses. Because a plausibly labelled attribution graph need not faithfully describe the underlying computation, independent reproduction on a different model tests whether the method's conclusions travel rather than merely re-confirming a single system.

This study independently reproduces that methodology in `Qwen3-4B-Instruct-2507`, an open-weight causal language model with 4.0 billion parameters, 36 transformer layers and hidden width 2,560 [7, 8]. The implementation uses small synthetic prompt sets, per-layer SAEs, pruned attribution-style graphs, and validation by inhibition or swap-in intervention. The principal behaviours are addition with emphasis on decimal carry, capital-city reasoning, and SI-unit reasoning. Addition and capital-city prompts parallel examples in the target study, while controlled force, energy and mass prompts extend the analysis into a scientific domain.

Three research questions organise the analysis:

1. Can a lightweight per-layer SAE pipeline identify compact feature sets whose interventions alter a pre-specified output contrast?
2. Do such effects generalise beyond the graph prompt and distinguish the hypothesised concept from matched alternatives?
3. When a sparse intervention fails, do reconstruction diagnostics and broader activation controls identify where the reproducibility chain breaks?

The structure of the analysis follows the evidence as it developed. Weak initial sparse interventions led to direct measurement of SAE density, which motivated a fixed TopK comparison. The revised representation yielded a compact and specific unit-domain effect. Initial arithmetic controls rejected graph-derived and ten-feature carry interpretations, while output-digit-balanced analysis subsequently produced a smaller 20-feature effect that survived a frozen case-level replication.

2 Background and relation to the target study

The following background differentiates the per-layer SAE decomposition used here from the target study's cross-layer replacement model and defines the quantities used in training, graph construction and causal intervention.

2.1 Sparse activation decompositions

At a fixed transformer layer and token position, let $x \in \mathbb{R}^d$ denote the output of the transformer's multilayer perceptron (MLP), where d is the model's hidden width, and let u denote the activation divided by the positive layer-specific scale defined in Section 3.2. An SAE with m latent coordinates uses encoder and decoder matrices $W_\mathrm{e} \in \mathbb{R}^{m\times d}$ and $W_\mathrm{d} \in \mathbb{R}^{d\times m}$, encoder bias $b_\mathrm{e} \in \mathbb{R}^m$, and decoder bias $b_\mathrm{d} \in \mathbb{R}^d$. It encodes u into $z \in \mathbb{R}^m_{\geq 0}$ and reconstructs it as $\hat{u} \in \mathbb{R}^d$:

$$z = \mathrm{ReLU}\left(W_\mathrm{e}(u - b_\mathrm{d}) + b_\mathrm{e}\right), \tag{1}$$

$$\hat{u} = W_\mathrm{d} z + b_\mathrm{d}. \tag{2}$$

Here $\mathrm{ReLU}(v)_i = \max(v_i, 0)$ acts coordinatewise. Since $m > d$, the code is overcomplete; sparsity rather than compression is intended to make each active z_i useful. Decoder column i of W_d is the activation-space direction associated with z_i, and all such columns form the learned dictionary. Reconstruction fidelity is measured by the fraction of variance explained,

$$\mathrm{FVE} = 1 - \frac{\sum_n \|u_n - \hat{u}_n\|_2^2}{\sum_n \|u_n - \bar{u}_\mathrm{train}\|_2^2}. \tag{3}$$

where n is the index of the validation example, $\bar{u}_\mathrm{train}$ is the coordinatewise training mean, and $\|\cdot\|_2$ is the Euclidean norm. FVE values one and zero respectively denote exact reconstruction and performance equal to the training-mean predictor. Note that high FVE does not establish monosemanticity (one coherent concept per coordinate), sparse causal control, or fidelity on artificial edits outside the clean-data distribution.

The follow-up models use a hard TopK activation,

$$z = \mathrm{TopK}_k(\mathrm{ReLU}(W_\mathrm{e}(u - b_\mathrm{d}) + b_\mathrm{e})), \tag{4}$$

where $\mathrm{TopK}_k(v)$ retains the k largest coordinates and zeros the rest, hence the number of non-zero coordinates is at most k. This replaces tuning an ℓ_1 mean-absolute-activation penalty [6]. Each decoder column is also projected to Euclidean norm one after every optimiser step, removing a scale ambiguity and expressing latent magnitudes along unit activation-space directions.

2.2 Anthropic's attribution graphs

The Anthropic paper used a 30-million-feature cross-layer transcoder (CLT) for Claude 3.5 Haiku rather than independent per-layer output SAEs [1]. In the following simplified notation, r is the index of a feature's source layer and ℓ of a reconstructed destination layer; $x^r \in \mathbb{R}^d$ is the layer-r residual stream (accumulated hidden state passed between transformer blocks), $a^r \in \mathbb{R}^{m_r}$ is its sparse feature vector, and $\hat{y}^\ell \in \mathbb{R}^d$ is the reconstructed layer-ℓ MLP output:

$$a^r = \mathrm{JumpReLU}(W_{\mathrm{enc}}^r x^r), \tag{5}$$

$$\hat{y}^\ell = \sum_{r \le \ell} W_{\mathrm{dec}}^{r \to \ell} a^r. \tag{6}$$

Here $W_{\mathrm{enc}}^r \in \mathbb{R}^{m_r \times d}$ maps x^r to feature preactivations, JumpReLU zeros values below a learned threshold, and $W_{\mathrm{dec}}^{r \to \ell} \in \mathbb{R}^{d \times m_r}$ maps a^r into layer-ℓ MLP space; omitted biases and thresholds do not alter the indexing. Because a^r may contribute to every $\ell \ge r$, cross-layer effects are trained jointly. Anthropic's prompt-local replacement adds the observed residual $y^\ell - \hat{y}^\ell$, where y^ℓ is the true MLP output, and fixes selected attention and normalisation values, reproducing the clean forward pass. Perturbation faithfulness measures whether this local approximation and the original model respond similarly when features change [2].

Table 1 states the correspondence precisely. Independent output SAEs and a CLT are architecturally distinct, and this distinction informs the interpretation of the force-associated and carry-associated panels as well as the null capital swaps.

Table 1: Methodological correspondence to the target study.

Component	Anthropic target study	Present study
Model	Claude 3.5 Haiku and a smaller research model	Open-weight `Qwen3-4B-Instruct-2507`
Dictionary	Joint CLT; 30 million features for Haiku	Seven independent MLP-output SAEs; 8,192 latents each
Training support	All model layers, jointly trained on large corpora	1,000 behaviour-specific final-token activations at layers 4, 8, 12, 16, 20, 24 and 28
Local graph	Explicit error nodes; selected operations frozen	Clean-value straight-through splice; gradients through the original model
Feature support	Features across layers and token positions	SAE features at the final prompt position
Validation	Constrained feature interventions and faithfulness tests	Error-preserving feature edits, complete-latent and raw-MLP controls

3 Methods

3.1 Model, prompts and activation capture

All experiments used `Qwen3-4B-Instruct-2507` in non-thinking mode with deterministic next-token evaluation. The model was loaded through Hugging Face Transformers with bfloat16 weights. Prompts were tokenised directly rather than wrapped in a chat template. Python, NumPy and PyTorch seeds were fixed to 787 for dataset order, train–validation splitting and initial SAE training. The selected transformer layers were $\{4, 8, 12, 16, 20, 24, 28\}$.

Three deterministic datasets were generated. The arithmetic dataset contains 1,000 two-digit sums, balanced over the 100 pairs of ones digits. The unit dataset contains five quantities, 20 contextual objects and ten templates, giving 1,000 prompts. The capitals dataset expands 189 country and state examples to 1,000 prompts by substituting non-capital cities from a fixed list. A forward hook recorded the MLP output at the final input token, producing one 1000×2560 matrix per selected layer and behaviour. Separate SAE sets were trained for arithmetic, units and capitals.

Baseline scoring used full greedy completions for addition and prefix-aware continuation for units and capitals. Causal interventions then evaluated the next-token distribution at a fixed prompt endpoint; capital swaps compared the first answer-token prefixes for source and target cities.

For input tokens $w_{1:n}$, let $h_n \in \mathbb{R}^d$ be the last-position hidden state, W_U the unembedding matrix, and b_U its bias, with $b_\mathrm{U} = 0$ for a bias-free output head. The next-token logits and probabilities are

$$\boldsymbol{\ell} = W_\mathrm{U} h_n + b_\mathrm{U}, \qquad p(q \mid w_{1:n}) = \frac{\exp(\ell(q))}{\sum_{v \in \mathcal{V}} \exp(\ell(v))}, \qquad \ell(q_1) - \ell(q_2) = \log \frac{p(q_1 \mid w_{1:n})}{p(q_2 \mid w_{1:n})}. \tag{7}$$

Thus a logit difference is the log probability ratio targeted by the contrast graphs and causal tests.

Though initial prediction was multi-token, arithmetic was evaluated autoregressively at the tens digit. For example,

```
Question: What is 58 + 83? Answer: 1
```

forces the hundreds digit of 141. The correct next token is `4` and the counterfactual produced by dropping the carry from $8 + 3$ is `3`. Matched no-carry sources preserved token length and the hundreds digit. The additional `44+83=127` source has next digit `2`, rather than the target's dropped-carry counterfactual `3`; transfer toward `2` therefore diagnoses source-digit copying, whereas a selective shift toward `3` is consistent with removal of the target's carry contribution.

Similarly, unit interventions were selected from known correct multi-token answers, but only operated on first-token answer prefixes. The principal output contrast is

$$G_\mathrm{unit} = \ell(\texttt{new}) - \ell(\texttt{j}), \tag{8}$$

where $\ell(q)$ is the pre-softmax next-token logit defined in Equation 7. The token `new` is the model's first token for "newtons", and `j` is its first token for "joules". A larger G_unit therefore means a stronger relative preference for newtons. Force and mass prompts act as activation donors to an energy target. The mass donor is a matched factual-recall control which changes the physical quantity but should not transfer the newtons preference.

3.2 SAE training and diagnostic selection

For each layer, a shared seeded 80:20 split supplied 800 training and 200 validation vectors. Activations were divided by

$$s_\ell = \frac{\mathbb{E}_\mathrm{train} \|x^\ell\|_2}{\sqrt{2560}}, \tag{9}$$

where x^ℓ is a training-set MLP output at layer ℓ and $\mathbb{E}_\mathrm{train}$ averages over the 800 training examples. Because $d = 2560$, this choice makes the mean Euclidean norm of $u^\ell = x^\ell / s_\ell$ approximately $\sqrt{d}$. Each SAE had input width 2,560 and latent width 8,192. The initial ReLU models were trained for 75 epochs with Adam, learning rate 10^{-3}, batch size 64 and

$$\mathcal{L}_\mathrm{ReLU} = \mathrm{mean}\left[(u - \hat{u})^2\right] + 3 \times 10^{-4}\, \mathrm{mean}(|z|). \tag{10}$$

where each mean is taken over the examples and coordinates in a minibatch, and $|z|$ is coordinatewise absolute value. The first term minimises reconstruction error, while the second regularises latent magnitude to encourage sparse activation. The checkpoint with minimum validation loss was retained. Each SAE contained 41,953,792 trainable parameters and was fitted to 800 training vectors.

After the initial interventions, all three ReLU sets were evaluated on their fixed validation splits using FVE, mean non-zero latent count ($L_0 = \|z\|_0$), cosine similarity between u and $\hat{u}$, and dead-feature fraction, defined as the fraction of latent coordinates that never activated on the evaluated set. The diagnostic motivated a pre-specified TopK comparison for arithmetic and units. Candidate values $k \in \{128, 256, 512\}$ were fixed before any candidate intervention result was examined. These models used the same layer set, width, data split and 75-epoch schedule, but Equation 4, unit-normalised decoder columns and no ℓ_1 term. A candidate was eligible only if mean validation FVE was at least 0.90, every layer's FVE was at least 0.85, and mean dead-feature fraction was at most 0.80. Among eligible candidates, the model with the smallest mean validation L_0 was selected. Intervention outcomes were not inputs to this selection rule.

3.3 Contrast attribution graphs

At graph construction time each SAE was inserted at the final prompt position with a clean-value straight-through splice. Writing $\hat{x} = s_\ell \hat{u}$ for the decoded activation in the model's original scale,

$$x' = x + \hat{x} - \text{stopgrad}(\hat{x}). \tag{11}$$

where stopgrad(v) returns v in the forward pass but has derivative zero. Consequently $x' = x$ numerically, while derivatives of downstream logits include the path through $\hat{x}$ and the SAE. This prevents reconstruction error from changing the baseline before features are attributed.

For target token t and contrast token c, the graph objective is

$$J = \ell(t) - \ell(c). \tag{12}$$

where t and c are vocabulary-token IDs, $\ell(\cdot)$ is the pre-softmax logit defined above, and positive J favours the target over the contrast. At the clean activation, the first-order attribution of active latent coordinate i is $A_i = z_i \, \partial J / \partial z_i$: its activation multiplied by the local derivative of the contrast with respect to that activation. For feature i at an earlier selected layer and feature j at the next selected layer, the edge quantity is

$$E_{i \to j} = z_i \frac{\partial z_j}{\partial z_i} \frac{\partial J}{\partial z_j}. \tag{13}$$

which multiplies the source activation, its local influence on the downstream feature, and the downstream feature's local influence on J. For prompt position p with embedding e_p, the input-token analogue is $e_p^\top \nabla_{e_p} J$. The graph retains the 20 active nodes of largest $|A_i|$ per selected layer and edges with $|E_{i \to j}| > 10^{-6}$. Decoder directions are multiplied by the model's unembedding matrix, the final linear map from hidden states to vocabulary logits, to obtain three diagnostic output-alignment labels.

Figure 1 summarises the experimental flow from captured MLP vectors through layerwise SAE dictionaries and contrast attribution to sparse interventions and broader causal controls.

Figure 2 is the one-prompt graph used in the initial tests. The balanced panel in Figure 7 was selected across all SAE latents, not from this graph.

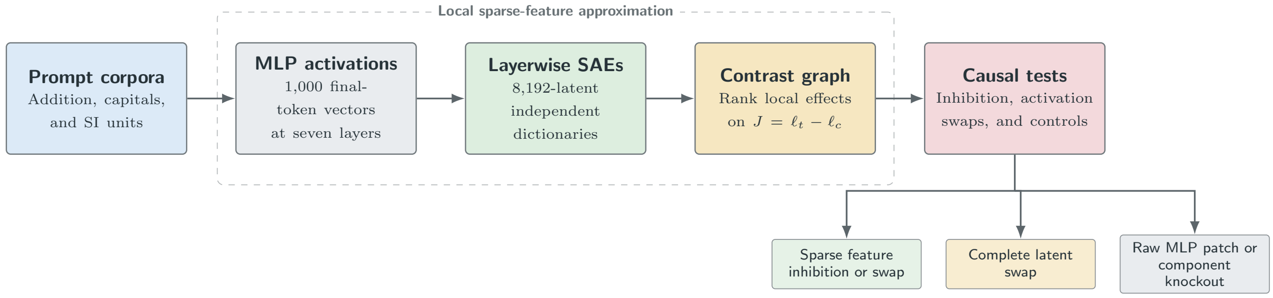

Figure 1: Independent small-scale reproduction pipeline. The lower branch distinguishes sparse feature tests from broader positive controls.

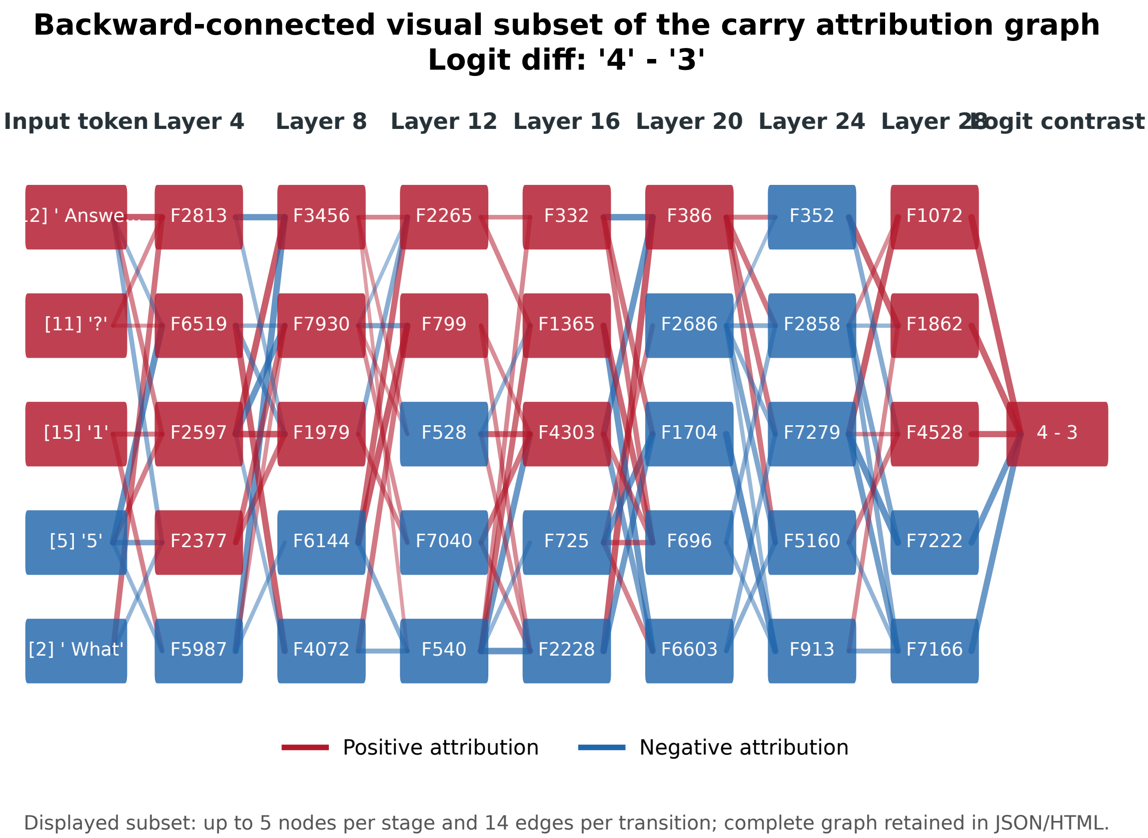

Figure 2: Backward-connected visual subset of the arithmetic attribution graph for $\ell(4) - \ell(3)$. Up to five nodes per stage are shown; the complete pruned graph is retained in JSON and interactive HTML form. Red and blue denote positive and negative local attribution respectively.

3.4 Causal interventions and confirmation protocols

Feature interventions preserve the clean reconstruction residual rather than replacing x by $\hat{x}$. Let S be the selected latent indices and let P_S be the diagonal projector that retains coordinates in S and zeros all others, so $z^{(-S)} = z - P_S z$. Inhibition edits the true MLP output by the decoded selected-feature difference alone,

$$x' = x + s_\ell W_\mathrm{d} \left(z^{(-S)} - z \right). \tag{14}$$

A source-to-target swap instead applies

$$x'_\mathrm{tgt} = x_\mathrm{tgt} + \alpha s_\ell W_\mathrm{d} P_S \left(z_\mathrm{src} - z_\mathrm{tgt} \right), \tag{15}$$

where x_tgt is the unedited target-prompt MLP output, z_src and z_tgt are source- and target-prompt codes at the same layer and token position, and α is the intervention strength. Selecting every coordinate, so $P_S = I$ for identity matrix I, swaps the complete 8,192-dimensional code. Raw MLP patching replaces target MLP outputs directly and bypasses the SAE. Component knockout sets an entire MLP, attention or transformer-block output to zero. These operations serve as positive and localisation controls; claims about sparse causal structure are based on selected-feature interventions.

Early diagnostics compared final-position and all-position edits. Token IDs and pre-edit MLP norms were printed to verify the actual positions and non-zero activation being modified. Because SAEs were trained at the final position, the final feature screens use only that position; all-position interventions are reported explicitly as distribution-mismatched exploratory controls.

A feature panel is a fixed set of SAE coordinates intervened on jointly; though it should be noted that the term does not imply monosemanticity. Two initial discovery-confirmation protocols were frozen. For arithmetic, features from one carry graph were ranked on eight discovery pairs by the target intervention effect minus the effect on a matched no-carry control. The predeclared top-ten panel was then tested on 24 disjoint pairs. Success required a negative mean carry-minus-control difference, a bootstrap 95% interval wholly below zero, and a negative mean carry-target effect. For units, 73 positive-attribution features from one force graph were ranked on eight physical systems by the force-source swap effect minus the matched mass-source effect. The frozen top-ten panel was tested on 16 disjoint systems. Success required both the force effect and force-minus-mass effect to be positive with bootstrap intervals wholly above zero. Each interval comprises the 2.5th and 97.5th percentiles of 5,000 seeded non-parametric resamples of evaluation cases and quantifies prompt-system variation, not model-sampling uncertainty. Capital experiments used graph-selected and complete-latent swaps on low- and high-confidence prompt pairs, without a separate feature-ranking protocol.

A final arithmetic protocol addressed a remaining confound: a feature may separate the correct and dropped-carry tokens because it encodes the output digit rather than carry status. For example, in `58+83`, a feature aligned with token `4` can distinguish the correct answer from `3` without representing the carry operation. After excluding 44 previously inspected arithmetic pairs, 105 fresh pairs met the clean-baseline criteria. Thirty-two pairs were assigned to discovery and 32 to confirmation while balancing shared tens-digit strata. At each layer, activation coordinates were standardised, the discovery mean for each output digit was subtracted, and the unit vector from the no-carry mean to the carry mean defined a linear raw-MLP score. Its conditioned area under the receiver-operating-characteristic curve (AUC) is the mean probability, within a shared output-digit stratum, that a carry case scores above a no-carry case. AUC values 0.5 and 1.0 respectively denote chance ranking and perfect separation.

All 57,344 TopK-256 latents were ranked on discovery data by their standardised carry-minus-no-carry activation difference within the same strata. Confirmation data did not affect ranking. The top-ten panel was the predeclared primary test, meaning that it alone determined whether the initial protocol succeeded; cumulative panels of 1, 3, 5 and 20 features were secondary panel-size diagnostics. The top-ten test did not meet the causal criterion, and the observed top-20 effect therefore remained a hypothesis-generating result. Its exact feature IDs were frozen in code and evaluated once on 32 of the 41 remaining baseline-qualified pairs that had received no feature intervention. Replication cases were selected using only baseline eligibility and output-digit balance. The stopping rule required a negative carry-target mean and a carry-minus-control bootstrap interval wholly below zero; no further panel or seed was permitted after this test.

The first units baseline-only screen used indirect wording and left too few fully qualified systems. Before any feature intervention was run, the protocol was amended to direct prompts of the form "the official SI unit used to measure [quantity] is named". Four wording variants were audited for 64 physical systems; eligible direct prompts were then split by system. The exact discovery and confirmation prompts were absent from the SAE corpus, and no physical system appeared in both splits. Recording this amendment establishes that prompt wording and eligibility were fixed before causal outcomes were observed.

3.5 Execution and reproducibility

The implementation is maintained in the public repository `https://gitlab.developers.cam.ac.uk/phy/data-intensive-science-mphil/assessments/projects/ep787`. Standalone Python modules under `src/` implement dataset generation, activation capture, SAE training, graph construction, intervention, diagnostics and figure generation; notebooks orchestrate these modules. YAML files record model and SAE settings, and all principal results are serialised as JSON or CSV before plotting.

GPU execution used Google Colab after CSD3 became unavailable during the experimental period. Each final notebook mounts Google Drive, pulls the current repository from GitHub, installs the package in editable mode, restores or trains checkpoints, and copies checkpoints and outputs back to Drive. Runs used NVIDIA T4, A100, and H100 GPUs. The model runs in bfloat16, whereas aggregate statistics and bootstrap calculations use NumPy precision. The complete notebook run order, standalone commands, configuration files and artifact locations are documented in `README.md`; earlier notebooks preserve exploratory provenance.

4 Results

4.1 Baseline competence

Table 2 reports the retained initial audit. Arithmetic was reliable, whereas units and capitals included multi-token answers and templates that sometimes elicited an intervening word rather than the answer prefix. Every causal benchmark therefore reapplied a clean-baseline eligibility rule to each specific prompt.

Table 2: Initial baseline audit. "In top-k" denotes an acceptable answer in the saved top-20 token list or prefix-aware completion check.

Behaviour	Prompts	Top-1 correct	In top-k
Addition	200	200 (100.0%)	200 (100.0%)
SI units	200	134 (67.0%)	193 (96.5%)
Capitals	189	94 (49.7%)	158 (83.6%)

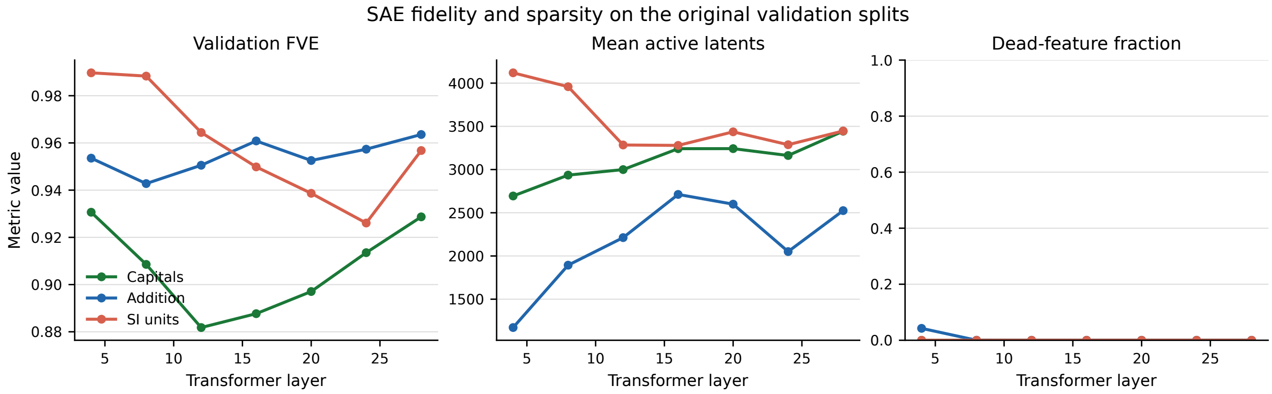

Figure 3: Layerwise fidelity and achieved sparsity of the original ReLU SAEs. High FVE coexists with thousands of active latents. The plotted dead-feature fraction is measured on combined train and validation support.

4.2 Original ReLU representation

The initial attribution graphs produced small sparse effects. Reconstruction diagnostics determined whether SAE error accounted for this result. Table 3 and Figure 3 show high fidelity: mean validation FVE was 0.954 for arithmetic, 0.959 for units and 0.907 for capitals, and mean cosine similarity was at least 0.993 in every domain. The corresponding codes nevertheless activated 2,167, 3,545 and 3,103 latents per example, approximately 26%, 43% and 38% of the dictionary. The original SAEs were therefore reconstructive but lacked a compact latent bottleneck.

Table 3: Original ReLU SAE diagnostics averaged over seven layers on the fixed 200-vector validation splits. Dead fraction is the fraction of dictionary features inactive on the validation split.

Domain	Mean validation FVE	Mean validation L_0	Mean dead fraction
Addition	0.9545	2,166.6	0.0106
SI units	0.9591	3,544.6	0.0000
Capitals	0.9068	3,102.8	0.0000

Broad interventions were introduced to test whether the model locations still contained causal information. For the unit minimal pair, an all-position complete-latent force-to-energy patch changed the top prefix from `j` to `new`: p(`new`) rose from 0.0240 to 0.4980 while p(`j`) fell from 0.7969 to 0.2354. The reverse complete-latent patch raised the `j` logit on the force target by 4.75 without changing its top token; broad transfer was therefore bidirectional in logits but asymmetric in answer. Swapping the 76 positive force-graph features into the energy target increased the `new` logit by 0.37 without changing the top token, while the reverse sparse graph swap was negligible. On six baseline-qualified graph-held-out pairs, final-position sparse swapping increased G_{unit} by a mean 0.229 logits (95% interval $[0.146, 0.333]$), with the predicted direction in all six cases and no top-token transfer. The desired force-to-energy transfer was therefore present in the complete SAE code but only weakly concentrated in the original graph-selected coordinates.

Arithmetic showed the same hierarchy. On `58+83`, inhibiting all 73 positive carry-graph features reduced the correct-minus-dropped-carry gap from 7.12 to 6.63. A strength-two complete-latent patch from `54+83` reduced it to 0.75, and a raw all-position MLP patch made `3` the top token. Under the same broad patch, an alternative `44+83` source made its own next digit, `2`, the top token, identifying source-digit transfer rather than an abstract carry state. On 12 graph-held-out carry pairs, raw MLP patching changed the gap by -13.99 logits and transferred the source top token in all cases, whereas ReLU sparse inhibition changed it by -0.396 ($[-0.458, -0.333]$). The raw control establishes causal sufficiency of the selected MLP state and source-digit dependence of the transferred information.

An all-position MLP knockout supplied a complementary diagnostic. Across the seven selected layers, `4` remained top but its probability fell from approximately 1.0 to 0.6406; `3` rose to 0.1836 and `2` to 0.1426. Layer 24 alone made `2` the top prediction. The simultaneous increase of several alternatives localises answer selection to layer 24 without isolating carry-specific computation. Attention knockout was weaker, while full-block knockout produced non-answer tokens and was excluded from concept-level interpretation. Since even the seven-layer MLP knockout preserved `4` as the top token, top-token transfer is a stringent endpoint in this example; subsequent tests therefore use matched changes in logit gaps as their primary causal estimand.

4.3 Capital-city controls

Capital-city experiments isolated clean prediction confidence as a potential source of weak transfer. In the initial Dallas/Oakland pair, `Austin` had probability 0.2773 for Dallas, while `Sacramento` had probability 0.2217 for Oakland and was not the top token. Complete-latent patches moved the relevant logits but did not exchange the answers. A second pair therefore used high-confidence baselines: Zarqa predicted the `Amman` prefix with probability 0.9844 and Basra predicted `Baghdad` with probability 0.9648. Patching Basra into Zarqa raised the `Baghdad` logit from 6.97 to 12.62, but its probability remained 0.0002 and `Amman` remained top. Graph-selected swaps were smaller; Figure 4 summarises both pairs.

The high-confidence control rules out baseline uncertainty as the sole source of weak transfer. Under the tested final-position MLP protocol, capital retrieval did not transfer. Representations in other components and the multi-token retrieval trajectory remain outside the measured intervention scope, so these experiments identify the boundary of the tested intervention without determining the cause of the null transfer.

4.4 Initial TopK arithmetic specificity tests

The ReLU diagnostic motivated a representation-level correction with an explicit support-size constraint. Arithmetic TopK candidates preserved high fidelity while reducing activity by approximately an order of magnitude. TopK-128 achieved mean FVE 0.9466 and mean $L_0 = 124.6$, but its mean dead-feature fraction of 0.832 exceeded the predeclared threshold. TopK-256 was therefore selected with mean FVE 0.9410, minimum layer FVE 0.9319, mean $L_0 = 171.6$ and dead fraction 0.716. TopK-512 was eligible but denser and was not selected.

On 12 graph-held-out all-position pairs, inhibiting the selected graph features reduced the carry-target

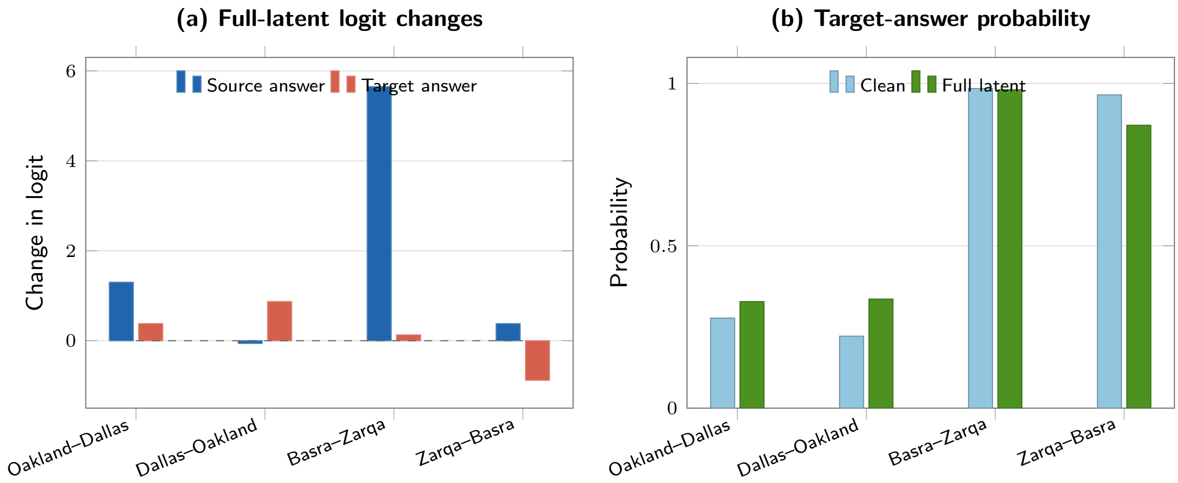

Figure 4: Complete-latent capital-city interventions. Increasing clean confidence from the Dallas/Oakland pair to Zarqa/Basra did not yield answer transfer.

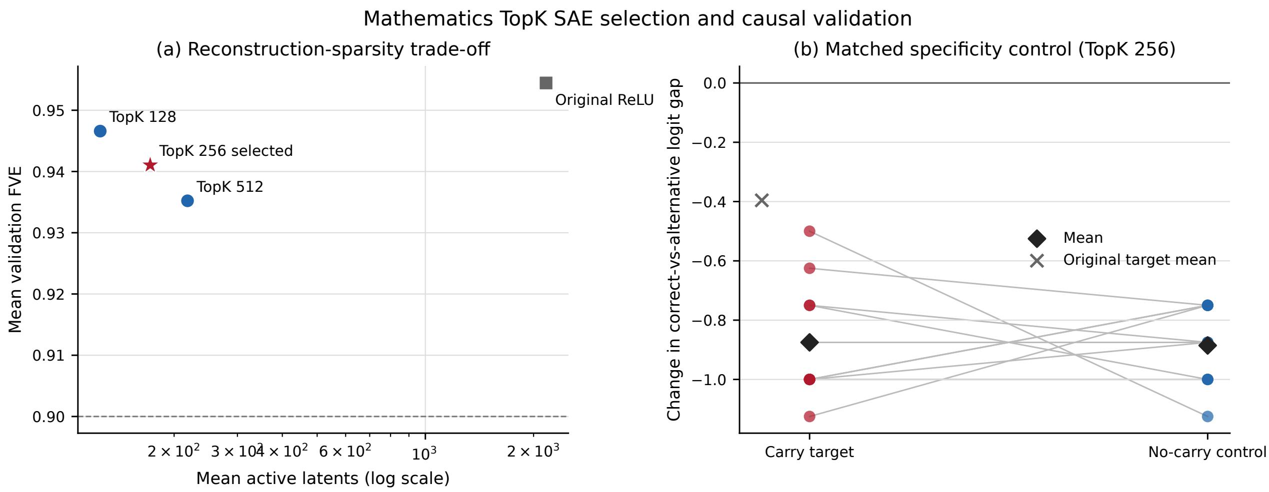

Figure 5: Predeclared arithmetic TopK selection and matched causal test. The selected representation is substantially sparser, but its graph-feature effect is equally present on no-carry controls. Lines in panel (b) connect matched prompt pairs.

gap by a mean -0.875 logits ($[-0.969, -0.771]$), more than twice the original ReLU mean. Every target moved in the expected direction. The matched no-carry gap, however, changed by -0.885 logits. The paired target-minus-control estimate was 0.010 ($[-0.125, 0.167]$), with the carry target more negative in only four of 12 pairs. The intervention therefore altered arithmetic digit confidence but was not selective for carrying, as visualised by the matched effects in Figure 5.

Position mismatch could have concealed a final-token feature effect. Repeating the benchmark at the SAE training position produced a mean target change of -0.656 and control change of -0.625; the paired estimate remained indistinguishable from zero at -0.031 ($[-0.156, 0.104]$). Finally, features were ranked on eight discovery pairs and a frozen top-ten panel was evaluated on 24 disjoint confirmation pairs. The target gap decreased by -0.141 logits, but the no-carry control also decreased by -0.109. The paired specificity effect was -0.031 ($[-0.089, 0.026]$); its interval crossed zero, so the predeclared criterion was not met. Twenty-one of 24 carry targets moved in the predicted direction, yet none changed top prediction. Five matched random panels produced paired means between -0.036 and

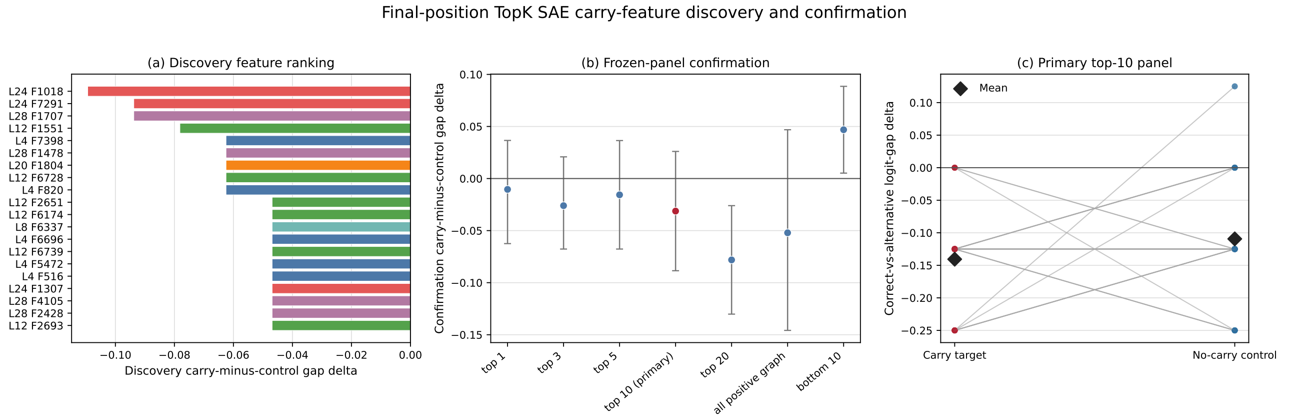

Figure 6: Final-position arithmetic discovery and frozen confirmation. Negative values favour carry selectivity. The primary top-ten interval crosses zero, so the panel does not satisfy the predeclared carry-specificity criterion.

0.021, reinforcing the conclusion that the residual top-ten effect was small relative to panel variability.

This sequence distinguishes sensitivity from mechanism. The TopK dictionary made interventions larger and the selected graph features consistently affected digit logits, while the matched control identified a generic arithmetic effect. The graph-derived panel is therefore characterised as arithmetic-supporting rather than carry-specific. Figure 6 records the null frozen top-ten test, which motivated the output-digit-conditioned search across the full SAE dictionary rather than a claim that carry information was absent or intrinsically inseparable.

4.5 Output-digit-balanced carry localisation

The initial graph screens motivated a broader hypothesis. Carry information could still be present in the SAE basis but masked by answer-digit variation or simply absent from the single attribution graph used for selection. The balanced protocol therefore searched all 57,344 latents using discovery activations rather than graph membership. Output-digit-conditioned raw-MLP probes confirmed that the selected site contained carry information. Confirmation AUC was 1.000 at both layers 24 and 28, compared with 0.754 and 0.800 at layers 4 and 8 and values below 0.5 at layers 12-20. The activation-ranked top-20 SAE panel comprised 11 layer-24 and nine layer-28 latents. On the first 32-pair confirmation set, its mean within-digit standardised activation difference was 0.967 ($[0.866, 1.077]$).

The predeclared top-ten primary was retained as a null result. Although its confirmation activation AUC was 1.000 and its mean conditioned difference was 1.130 ($[1.011, 1.253]$), inhibition changed the carry-target gap by -0.008 logits and the no-carry gap by -0.023. The paired specificity estimate was therefore $+0.016$ ($[-0.035, 0.063]$), separating held-out decodability from causal selectivity.

The secondary top-20 panel produced a different pattern on the same confirmation set: carry-target inhibition changed the gap by -0.078 logits, the no-carry control changed it by $+0.023$, and the paired effect was -0.102 ($[-0.148, -0.051]$). Its secondary status defined this estimate as hypothesis-generating. The exact 20 IDs were then frozen without reranking and tested once on 32 intervention-untouched eligible pairs. In this case-level replication, the carry-target mean was -0.059 ($[-0.090, -0.023]$), the no-carry mean was $+0.027$ ($[-0.004, 0.059]$), and the paired specificity effect was -0.086 ($[-0.137, -0.039]$). The criterion passed; 62.5% of pairs had a more negative carry effect than control, and neither condition

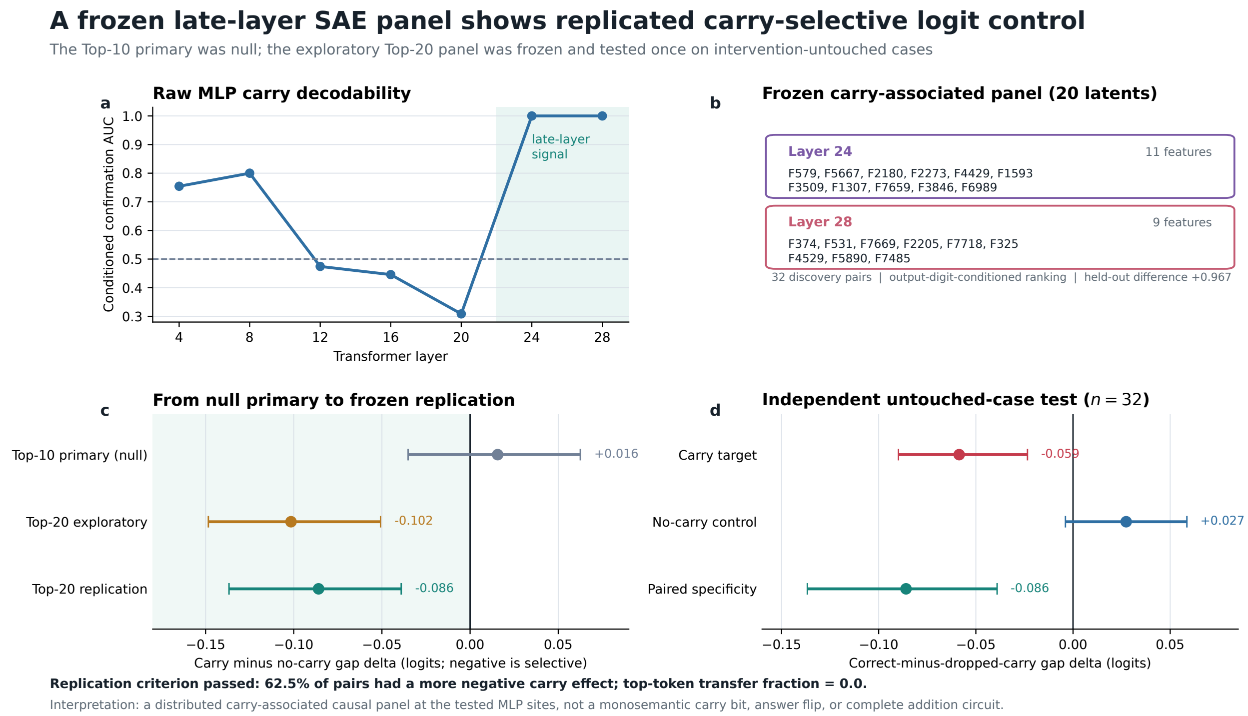

Figure 7: Output-digit-balanced arithmetic localisation and case-level replication. Panel (a) reports confirmation AUC for carry directions fitted in raw MLP outputs after output-digit centring. Panel (b) lists the exact activation-ranked top-20 carry-associated panel and the final-token inhibition applied to it. Panel (c) retains the null top-ten primary, distinguishes the secondary top-20 observation from its untouched-case replication, and reports paired carry-minus-control effects. Panel (d) decomposes the replication into target and matched-control changes. Error bars are seeded bootstrap 95% intervals across prompt pairs.

changed the top prediction.

Figure 7 establishes a carry-associated causal effect for a distributed late-layer SAE panel at the tested MLP sites. Inhibition selectively reduced preference for the carry-consistent tens digit relative to a matched no-carry control, and the effect recurred on cases excluded from feature ranking and the first intervention test. The measured scope is a small logit effect under one SAE basis, without output-token transfer or evidence of a single carry bit or complete arithmetic circuit.

4.6 Compact units panel

The same fixed TopK procedure was applied to units without changing the original SAE corpus. All three candidates met the diagnostic thresholds. TopK-128 was selected because it had the smallest mean validation activity: FVE 0.9358, minimum layer FVE 0.8891, mean $L_0 = 99.8$ and dead fraction 0.799. TopK-256 and TopK-512 had mean L_0 values 132.7 and 165.5 respectively. Relative to the original units ReLU SAE, the selected code used 35.5 times fewer active latents while losing only 0.0233 mean FVE.

An initial graph-held-out benchmark provided an independent directional check. Eight of 16 candidate prompt pairs met both clean source and target criteria. Across those eight, swapping all positive graph features at the final token increased G_{unit} by 0.586 logits ($[0.352, 0.883]$), with the predicted direction in every case. The complete latent and raw MLP controls produced changes of 1.352 and 1.617 logits.

No condition changed the top prediction. Compared with the original ReLU sparse effect of 0.229 on its six eligible pairs, TopK provided a larger but still broad graph-set effect.

The frozen feature screen then tested whether that effect could be concentrated and distinguished from another unit donor. Of the 73 positive graph features, the ten highest-ranked discovery features comprised eight layer-28 features, one layer-8 feature and one layer-4 feature. Six decoder directions had force-related top-output labels, providing an output-alignment diagnostic. Figure 8 places the frozen force panel in its source graph. Its eight layer-28 coordinates have retained direct output edges; two earlier coordinates lie at layers 4 and 8. On 16 disjoint confirmation systems, the force-source swap increased the newtons-minus-joules gap by 1.234 logits ($[1.148, 1.332]$). The mass-source control changed it by -0.090 ($[-0.129, -0.051]$), giving a paired specificity effect of 1.324 ($[1.223, 1.430]$). All 16 systems had a positive force effect and a force effect larger than the corresponding mass effect.

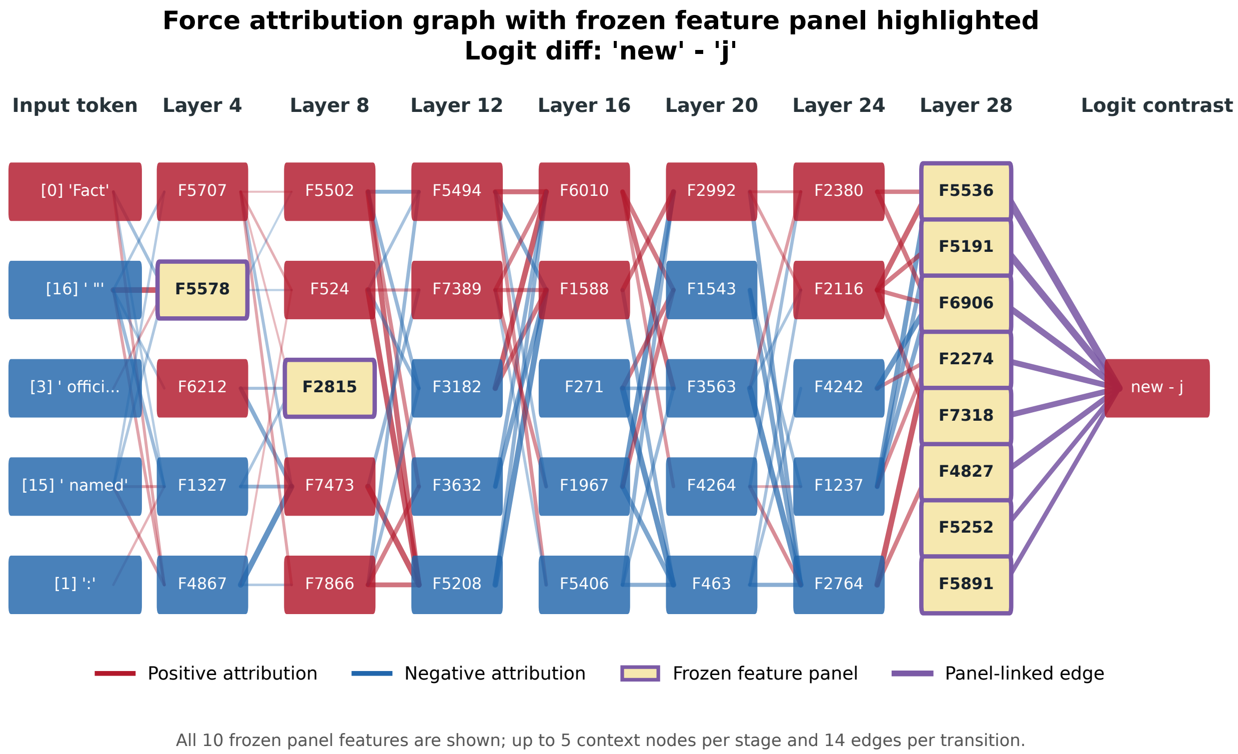

Figure 8: Force attribution graph with the frozen ten-feature panel highlighted. Yellow boxes mark selected coordinates; red and blue context encodes signed local attribution for $\ell(\texttt{new}) - \ell(\texttt{j})$. Purple lines retain graph edges linking selected features to each other or the output. All panel nodes are shown; other context is pruned.

The cumulative panel response grew monotonically with panel size. Force-minus-mass effects were 0.352, 0.813, 1.055 and 1.324 logits for the top 1, 3, 5 and 10 features. The top-20 panel gave 1.348 and all 73 positive graph features gave 1.391, whereas the reverse-ranked bottom-ten control gave only 0.055 ($[0.008, 0.098]$). Five layer-count-matched random panels had effects between 0.328 and 1.191; the primary panel exceeded all five, although five draws are insufficient for a formal randomisation test. The complete-latent and raw-MLP paired effects were 2.840 and 2.891 logits respectively.

In Figure 9, the clean mean confirmation gap was -9.41, the force-swapped gap was -8.17 and the mass-control gap was -9.50. Joules consequently remained the top prediction in every case. The force-source effect of the top-ten panel was 97.5% of the corresponding 73-feature force-source effect

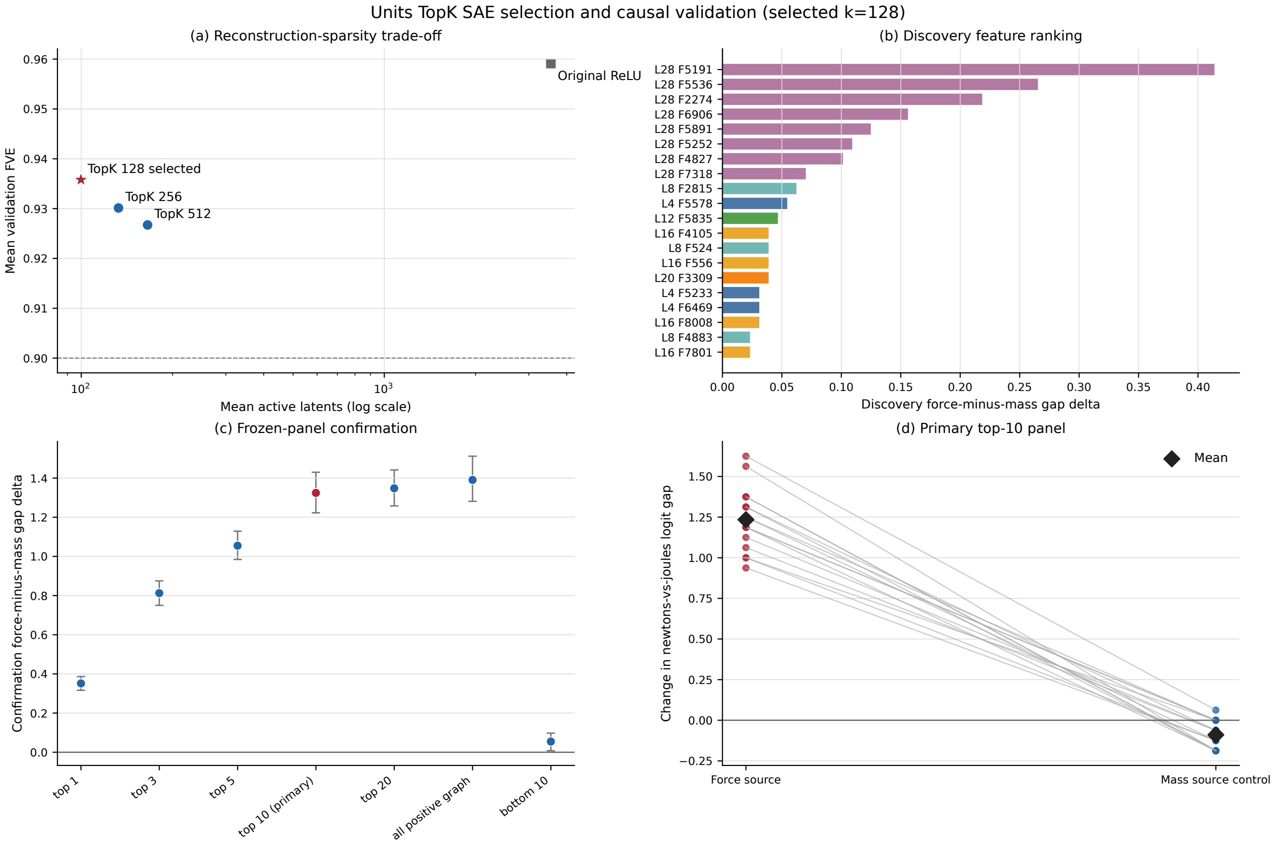

Figure 9: Units TopK selection, discovery ranking and frozen confirmation. Panel (a) shows the reconstruction-sparsity criterion that selected TopK-128 before intervention. Panel (b) ranks graph features by the discovery force-minus-mass effect, with the first ten forming the frozen panel. Panel (c) compares cumulative panel sizes with the full positive graph and reverse-ranked control. Panel (d) shows paired confirmation effects for all 16 systems; every force-source shift exceeds its mass-source control.

and 40.6% of the complete-latent force-source effect. These are ratios of observed intervention effects, not fractions of variance or of the mechanism explained.

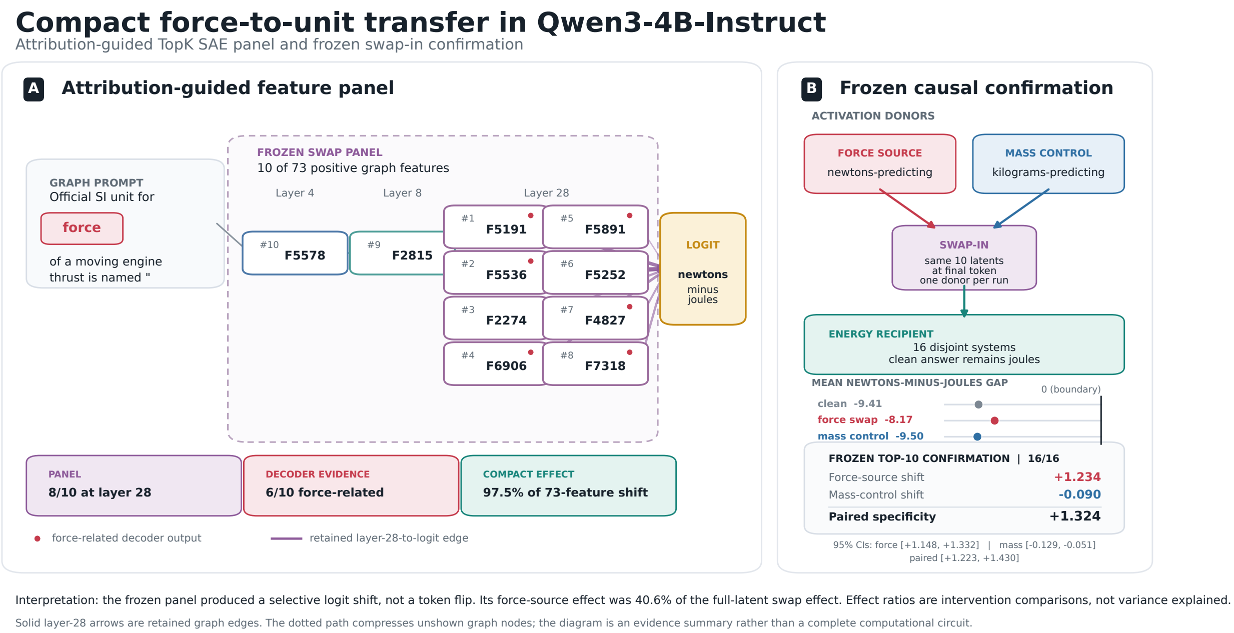

Figure 10: Evidence summary for the compact force-associated panel. All ten selected feature IDs and their layer assignments are shown; red markers identify the six features with force-related decoder-output labels, and solid purple lines are retained layer-28-to-logit graph edges. Swaps replace these ten final-token latent values with force- or mass-source values. The measured outcome is a selective logit shift, and the diagram represents the tested feature panel rather than a complete computational circuit.

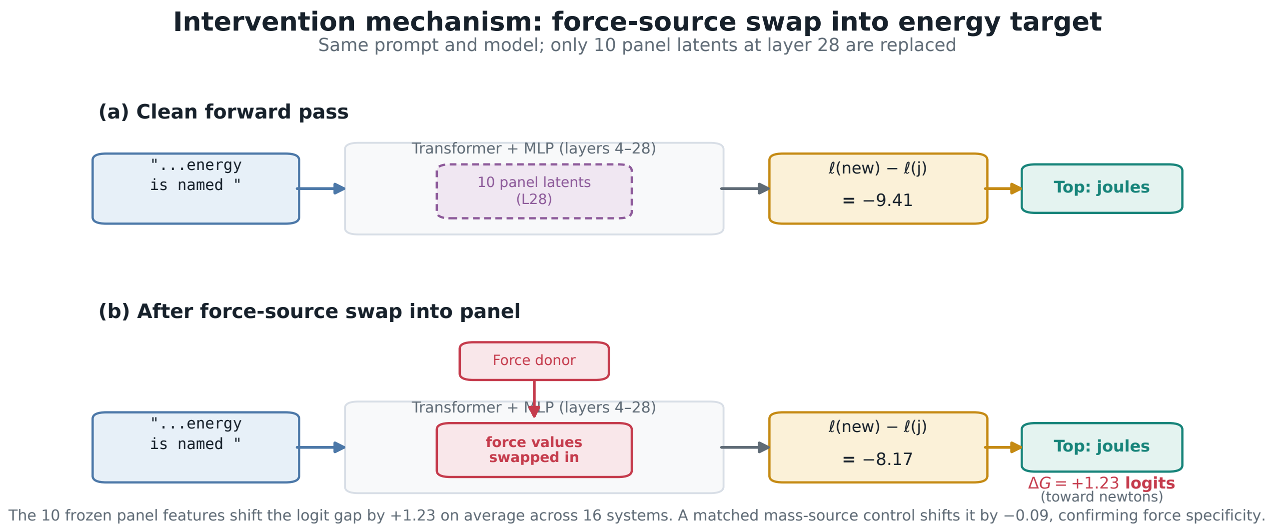

Figure 11: Schematic of the panel swap intervention. Panel (a) shows the clean energy-target forward pass, where the model strongly predicts joules. Panel (b) shows the same prompt after replacing only the 10 frozen panel latents at layer 28 with values from a force-predicting source. The newtons-minus-joules logit gap shifts by +1.23 on average, while the top prediction remains joules.

Figure 10 integrates panel membership, retained graph relations and the force-versus-mass comparison. Figure 11 isolates the intervention operation: force-source values replace the same ten coordinates while the energy target remains fixed. The result satisfies the predeclared causal criterion and identifies a small, attribution-guided, layer-28-dominated group that carries force-associated information and selectively shifts unit-token preference across new physical systems. Its measured scope is causal sufficiency for the logit shift; unique necessity, coordinate-level monosemanticity and complete factual-recall computation were not tested.

5 Discussion

5.1 Reproduction outcomes

The study reproduced the complete small-scale workflow, including fixed behavioural baselines, MLP activation capture, behaviour-specific per-layer SAEs, a pruned input-feature-logit graph, and causal validation in the underlying model. It recovered two sparse effects with different inferential histories. The units top-ten panel was selected under a predeclared protocol and generalised from one graph prompt through a separate discovery set to 16 disjoint confirmation systems, where it was specific relative to a mass donor. In arithmetic, the predeclared top-ten primary was null, while an activation-ranked top-20 secondary hypothesis subsequently reproduced under a frozen one-shot test on 32 intervention-untouched cases.

Relative to the target paper's circuit narratives, both results operate at the level of logit contrasts without answer changes. The experiments use one SAE seed, the arithmetic panel follows a null smaller primary, and the five units random panels provide a limited empirical null distribution. Accordingly, the results identify carry-associated and force-associated panels rather than complete circuits.

Arithmetic combines controlled null results with a replicated positive effect. Graph-derived features influenced digit logits after TopK retraining but did not distinguish carry targets from no-carry controls, while broad patches transferred source-specific answer state. Output-digit conditioning then isolated a late-layer signal whose 20-feature causal effect replicated across new cases. Concept-matched controls and held-out replication determine the resulting semantic interpretation. Capitals remained a null reproduction after clean confidence was increased, establishing domain variation in factual-recall transfer.

5.2 Why reconstruction was insufficient

The diagnostics identify three sources of divergence between reconstruction quality and causal feature control. First, the original ℓ_1 objective is scale-degenerate when decoder norms are unconstrained. For $0 < c < 1$, the transformation

$$W'_{\mathrm{e}} = cW_{\mathrm{e}}, \quad b'_{\mathrm{e}} = cb_{\mathrm{e}}, \quad W'_{\mathrm{d}} = c^{-1}W_{\mathrm{d}} \tag{16}$$

gives $z' = cz$ because $\mathrm{ReLU}(cv) = c\,\mathrm{ReLU}(v)$ for $c > 0$, and leaves $W'_{\mathrm{d}}z' = W_{\mathrm{d}}z$ unchanged while multiplying the latent ℓ_1 penalty by c. The raw penalty is therefore not scale-invariant. The observed L_0 values show its practical consequence: high FVE coincided with dense codes. TopK plus decoder normalisation removes this ambiguity by controlling support size directly, and the units result improved accordingly.

Second, reconstruction, decodability and causal separability are different objectives. Equation 3 constrains average clean-distribution error $\|x - \hat{x}\|$, whereas a latent edit Δz produces the decoded activation change $\Delta x = W_{\mathrm{d}}\Delta z$. For output objective J, a local expansion is

$$J(x + \Delta x) - J(x) = \nabla J(x)^{\top}\Delta x + O(\|\Delta x\|_2^2). \tag{17}$$

where $\nabla J(x)$ is the gradient and $O(\|\Delta x\|_2^2)$ collects quadratic and higher-order terms. Low reconstruction error does not bound these terms or require semantically distinct factors to occupy distinct coordinates. Correlated latents can reconstruct well while finite ablation or swap effects are distributed across many directions. The balanced arithmetic experiment illustrates this gap concretely. After conditioning on the output digit, the top-ten activation score classified held-out carry status perfectly, yet inhibiting those same ten features produced no selective causal effect. Selective control appeared only after expanding to 20 frozen latents, and it survived a separate case-level test.

Third, the independent per-layer output SAEs represent cross-layer effects differently from Anthropic's CLT. A CLT assigns one upstream feature decoder contributions to several later MLPs, whereas the present graph links separately trained dictionaries through local derivatives. The clean-value splice preserves the unperturbed output without explicit error nodes, and perturbation faithfulness remains an empirical property of the resulting graph [2].

5.3 Statistical and representational scope

Each behaviour-specific SAE set (approximately 42 million parameters per layer) was trained on 800 vectors drawn from a single templated prompt generator. The validation set shares the generator, so FVE characterises within-domain reconstruction rather than broad-language generalisation. This is appropriate for the project's scope, which targets a specific reproducibility question about identified behaviours rather than dictionary-level universality. Units confirmation prompts were absent from the

SAE training corpus but belong to the same physical-quantity domain.

All interventions operate at the final token of the MLP output. Claims about mechanism presence or absence are restricted to this component and position, and do not extend to attention heads or residual-stream features. The final screens match the SAE training position, and the intervention framework applies the same error-preserving formulation regardless of whether the prompt is a clean-baseline continuation or a teacher-forced digit state.

Confirmation sample sizes are 24 pairs (graph-derived arithmetic screen), 32 pairs (each balanced arithmetic set) and 16 unit systems. Bootstrap intervals reflect prompt-system variation and condition on the constructed evaluation pools. They do not account for SAE-seed variation, which is the first identified extension priority. The five layer-count-matched random panels in the units experiment provide a directional empirical null but are too few for a formal randomisation test. For arithmetic, the replication tests persistence of the exact selected panel on cases that had no prior feature intervention; testing across independently trained dictionaries remains the priority next step. Because the model uses bfloat16, individual logit values are quantised to approximately 0.0625 increments. The reported effects are aggregated across multiple cases and replications, so the conclusions do not rest on single-prompt logit differences.

5.4 Computational constraints and experimental coverage

CSD3, the intended high-performance computing environment, became unavailable during an extended cooling failure in late June 2026 [11]. CSD3's high-performance working storage (`hpc-work`) is explicitly documented as non-backed-up, and node-local scratch is purged between jobs [12]. The project's activation matrices, SAE checkpoints and intermediate outputs were lost and had to be regenerated from source code and the original model weights.

The replacement Google Colab environment provided paid GPU access (T4, A100, and H100 hardware) under a finite compute-unit allocation. Colab's resource model allocates units dynamically at a rate of approximately 6.77 units per GPU-hour, giving roughly 89 GPU-hours from the available 600-unit balance [9]. Model weights could not be uploaded as persistent storage due to file-size constraints, requiring re-download at the start of each session and adding significant overhead per run. The slower iterative cycle relative to a persistent cluster environment meant that experimental decisions had to be more targeted: compute was allocated to the predeclared candidate sweeps, held-out controls and frozen confirmations rather than exploratory multi-seed or all-layer investigations.

As a concrete consequence, the original ReLU SAE diagnostics, the TopK candidate training, and the attribution graphs all had to be recomputed from scratch after the CSD3 incident. Each full behaviour-specific TopK sweep (training three candidate SAEs across seven layers, building the attribution graph, and running the feature screen) required approximately four hours of continuous A100 time. The total regeneration cost was approximately 20 GPU-hours, which accounts for roughly a quarter of the available budget and directly constrained the number of SAE seeds and additional conditions that could be explored.

Both final TopK protocols used the same repository, configuration format and intervention framework, supporting direct comparison between arithmetic and units despite the environment change.

5.5 Prioritised further work

The most informative next experiment is replication of the units top-ten effect across SAE and graph seeds with an enlarged, pre-generated confirmation pool. The complete selection procedure, including feature ranking and panel size, must be rerun independently; retaining the same feature IDs after retraining would not test basis stability. More layer-count-matched random panels would permit a formal randomisation comparison.

For arithmetic, the next priority is a full repetition of output-digit-conditioned discovery and exact top-20 replication across independently trained SAE seeds, with feature ranking rerun from the beginning and a newly generated confirmation pool. Training SAEs on the forced decimal positions used for intervention, with carry status, source next digit and target next digit varied factorially, would test representational stability across learned bases.

At greater computational scale, per-layer transcoders or a small CLT should be compared with the independent output SAEs at matched reconstruction and active-feature budgets. Explicit error nodes and perturbation-faithfulness tests would align the graph semantics more closely with the target work. Attention or residual-stream decompositions provide a further test of factual retrieval mechanisms.

6 Conclusion

This study independently reproduced a small-scale circuit-tracing workflow for an open language model and evaluated each link in the inferential chain. The original ReLU SAEs achieved high held-out reconstruction fidelity but activated thousands of latents. Broad patches established that the selected MLP states contained causal answer information, while matched controls exposed source-digit transfer in arithmetic and weak capital-city transfer. A predeclared TopK correction substantially improved sparsity and intervention magnitude. The first arithmetic graph and top-ten tests remained non-specific; an output-digit-balanced extension subsequently identified a frozen 20-feature panel whose small carry-selective inhibition effect reproduced on 32 untouched cases. The units analysis independently yielded a stronger frozen ten-feature panel whose force-source effect generalised to 16 disjoint systems and was absent under a mass-source control.

The study provides a small-scale methodological reproduction with domain-dependent causal results. Sparse SAE features provide compact causal control in this Qwen setting, while reconstruction quality, decodability and directional logit movement measure distinct properties. Mechanistic interpretation follows from concept-matched controls, held-out confirmation, pre-specified handling of adaptive hypotheses, and separation of sparse interventions from broad activation patching. The units panel satisfies a predeclared criterion for a force-associated shift, the arithmetic panel satisfies a frozen case-level replication criterion for a smaller distributed carry-associated shift, and the null capital result establishes a domain boundary of transfer. Together, these positive and null findings constitute the principal reproducibility result.

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Declaration of use of autogeneration tools

Generative AI tools were used during software development to assist with tasks that do not require significant project-related decisoin making, such as code review, debugging, experimental documentation, figure generation, and proofreading of report. All meaningful parts of the study were conducted by Evelyn Paskhaver, including developing and running all code, and writing the report.